Penetration Testing Task (Analyze)	Answer Hint
1) Analyze a target network to determine whether hosts are alive without using ICMP packets.	Use TCP SYN/ACK or ARP discovery because ICMP may be blocked.

Aim

To identify active hosts in a target network without using ICMP packets by using TCP SYN and ARP-based host discovery techniques.

Theory

Host discovery is the process of identifying live or active devices on a network. A common method is using ICMP Echo requests, but ICMP may be blocked by firewalls or security devices. In such cases, TCP SYN/ACK probes or ARP requests can be used to identify active hosts.

A **TCP SYN discovery** sends SYN packets to selected TCP ports such as 80 and 443. A response from the target indicates that the host is reachable.

ARP discovery is mainly used within a local network. It sends ARP requests to determine whether a device with a particular IP address is present. ARP discovery is generally reliable on a local LAN because ARP operates at the data-link layer.

Tools Required

- Kali Linux
- Nmap
- Authorized target network

Procedure

1. Open the Kali Linux terminal.
2. Identify the authorized target network.
3. Perform host discovery using TCP SYN probes.

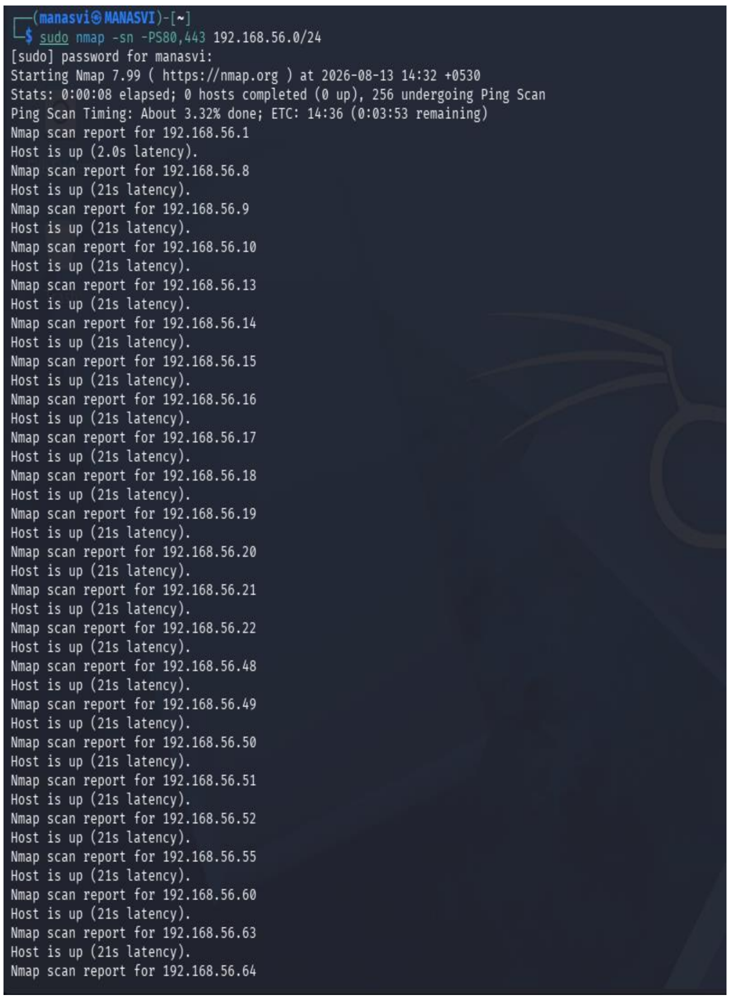
4. Perform ARP-based host discovery if the target is on the same local network.

5. Observe the Nmap output and identify hosts reported as **Host is up**.

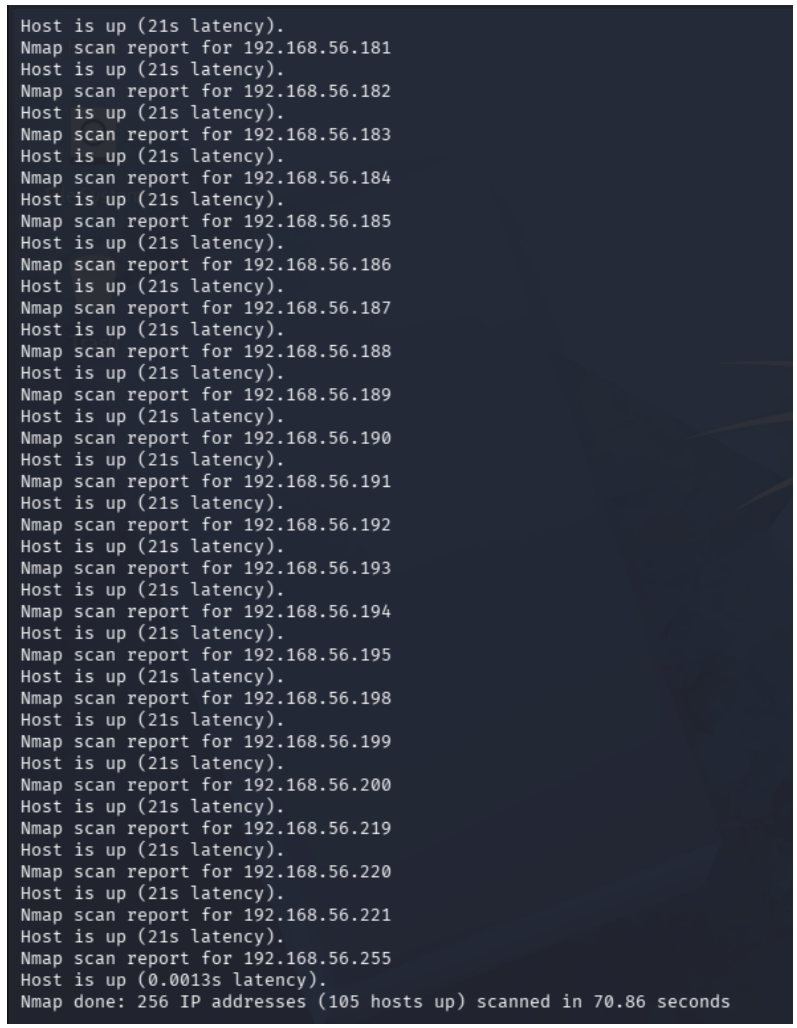
6. Compare the results obtained using the two techniques.

Command 1 – TCP SYN Host Discovery

sudo nmap -sn -PS80,443 192.168.56.0/24



```
┌──(manasvi㉿MANASVI)-[~]
└─$ sudo nmap -sn -PS80,443 192.168.56.0/24
[sudo] password for manasvi:
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-13 14:32 +0530
Stats: 0:00:08 elapsed; 0 hosts completed (0 up), 256 undergoing Ping Scan
Ping Scan Timing: About 3.32% done; ETC: 14:36 (0:03:53 remaining)
Nmap scan report for 192.168.56.1
Host is up (2.0s latency).
Nmap scan report for 192.168.56.8
Host is up (21s latency).
Nmap scan report for 192.168.56.9
Host is up (21s latency).
Nmap scan report for 192.168.56.10
Host is up (21s latency).
Nmap scan report for 192.168.56.13
Host is up (21s latency).
Nmap scan report for 192.168.56.14
Host is up (21s latency).
Nmap scan report for 192.168.56.15
Host is up (21s latency).
Nmap scan report for 192.168.56.16
Host is up (21s latency).
Nmap scan report for 192.168.56.17
Host is up (21s latency).
Nmap scan report for 192.168.56.18
Host is up (21s latency).
Nmap scan report for 192.168.56.19
Host is up (21s latency).
Nmap scan report for 192.168.56.20
Host is up (21s latency).
Nmap scan report for 192.168.56.21
Host is up (21s latency).
Nmap scan report for 192.168.56.22
Host is up (21s latency).
Nmap scan report for 192.168.56.48
Host is up (21s latency).
Nmap scan report for 192.168.56.49
Host is up (21s latency).
Nmap scan report for 192.168.56.50
Host is up (21s latency).
Nmap scan report for 192.168.56.51
Host is up (21s latency).
Nmap scan report for 192.168.56.52
Host is up (21s latency).
Nmap scan report for 192.168.56.55
Host is up (21s latency).
Nmap scan report for 192.168.56.60
Host is up (21s latency).
Nmap scan report for 192.168.56.63
Host is up (21s latency).
Nmap scan report for 192.168.56.64
```



```
Host is up (21s latency).
Nmap scan report for 192.168.56.181
Host is up (21s latency).
Nmap scan report for 192.168.56.182
Host is up (21s latency).
Nmap scan report for 192.168.56.183
Host is up (21s latency).
Nmap scan report for 192.168.56.184
Host is up (21s latency).
Nmap scan report for 192.168.56.185
Host is up (21s latency).
Nmap scan report for 192.168.56.186
Host is up (21s latency).
Nmap scan report for 192.168.56.187
Host is up (21s latency).
Nmap scan report for 192.168.56.188
Host is up (21s latency).
Nmap scan report for 192.168.56.189
Host is up (21s latency).
Nmap scan report for 192.168.56.190
Host is up (21s latency).
Nmap scan report for 192.168.56.191
Host is up (21s latency).
Nmap scan report for 192.168.56.192
Host is up (21s latency).
Nmap scan report for 192.168.56.193
Host is up (21s latency).
Nmap scan report for 192.168.56.194
Host is up (21s latency).
Nmap scan report for 192.168.56.195
Host is up (21s latency).
Nmap scan report for 192.168.56.198
Host is up (21s latency).
Nmap scan report for 192.168.56.199
Host is up (21s latency).
Nmap scan report for 192.168.56.200
Host is up (21s latency).
Nmap scan report for 192.168.56.219
Host is up (21s latency).
Nmap scan report for 192.168.56.220
Host is up (21s latency).
Nmap scan report for 192.168.56.221
Host is up (21s latency).
Nmap scan report for 192.168.56.255
Host is up (0.0013s latency).
Nmap done: 256 IP addresses (105 hosts up) scanned in 70.86 seconds
```

Explanation:

- sudo – executes Nmap with administrator privileges.

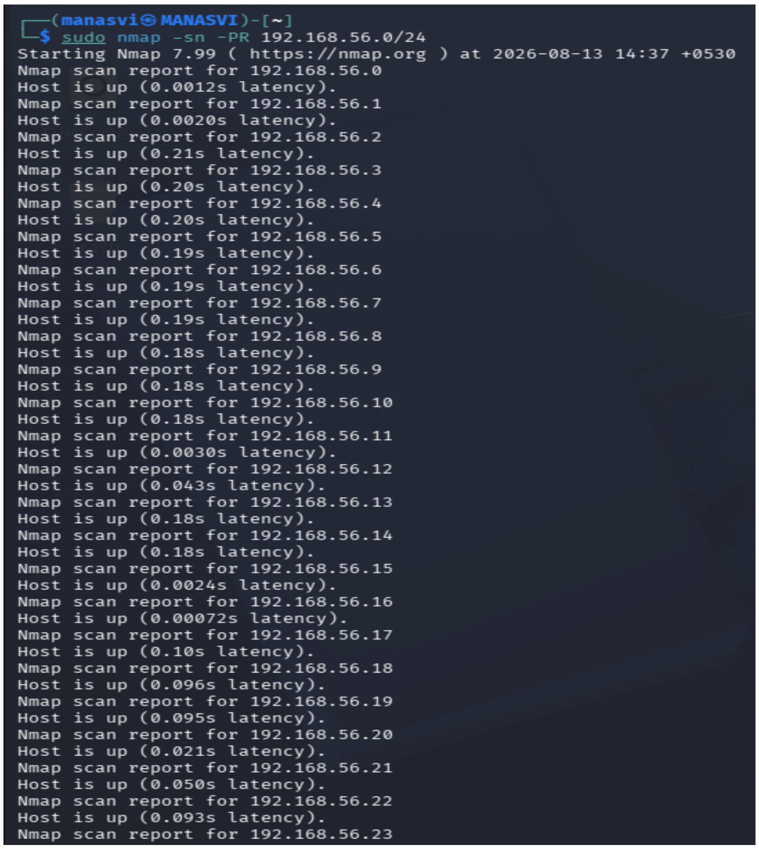
- nmap – network scanning tool.

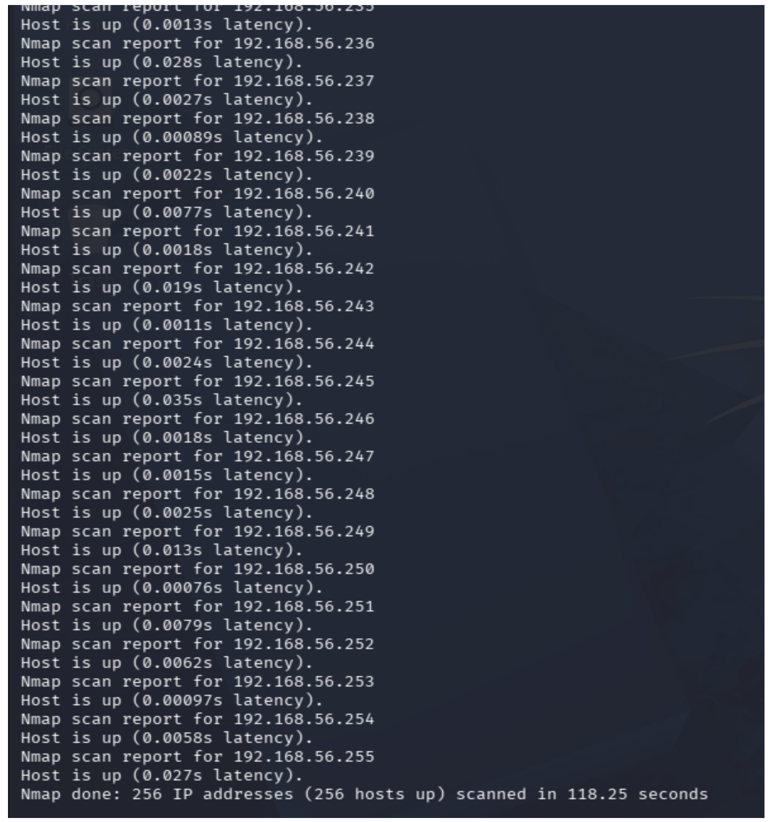
- -sn – performs host discovery without a port scan.

- -PS80,443 – uses TCP SYN probes against ports 80 and 443.
- 192.168.56.0/24 – target network

Command 2 – ARP Host Discovery

sudo nmap -sn -PR 192.168.56.0/24





Explanation:

- -PR – performs ARP discovery.
- It is particularly suitable when scanning hosts on the same local network.

Result

The target network was successfully analyzed to identify live hosts without relying on ICMP packets. **TCP SYN and ARP discovery** provided alternative methods for host discovery when ICMP may be blocked.

Conclusion

TCP SYN/ACK and ARP discovery are useful alternatives to ICMP-based host discovery. ARP discovery is particularly effective on a local LAN, while TCP-based discovery can be useful when ICMP is blocked but TCP traffic is permitted. These techniques help security testers identify active hosts during an authorized penetration test.